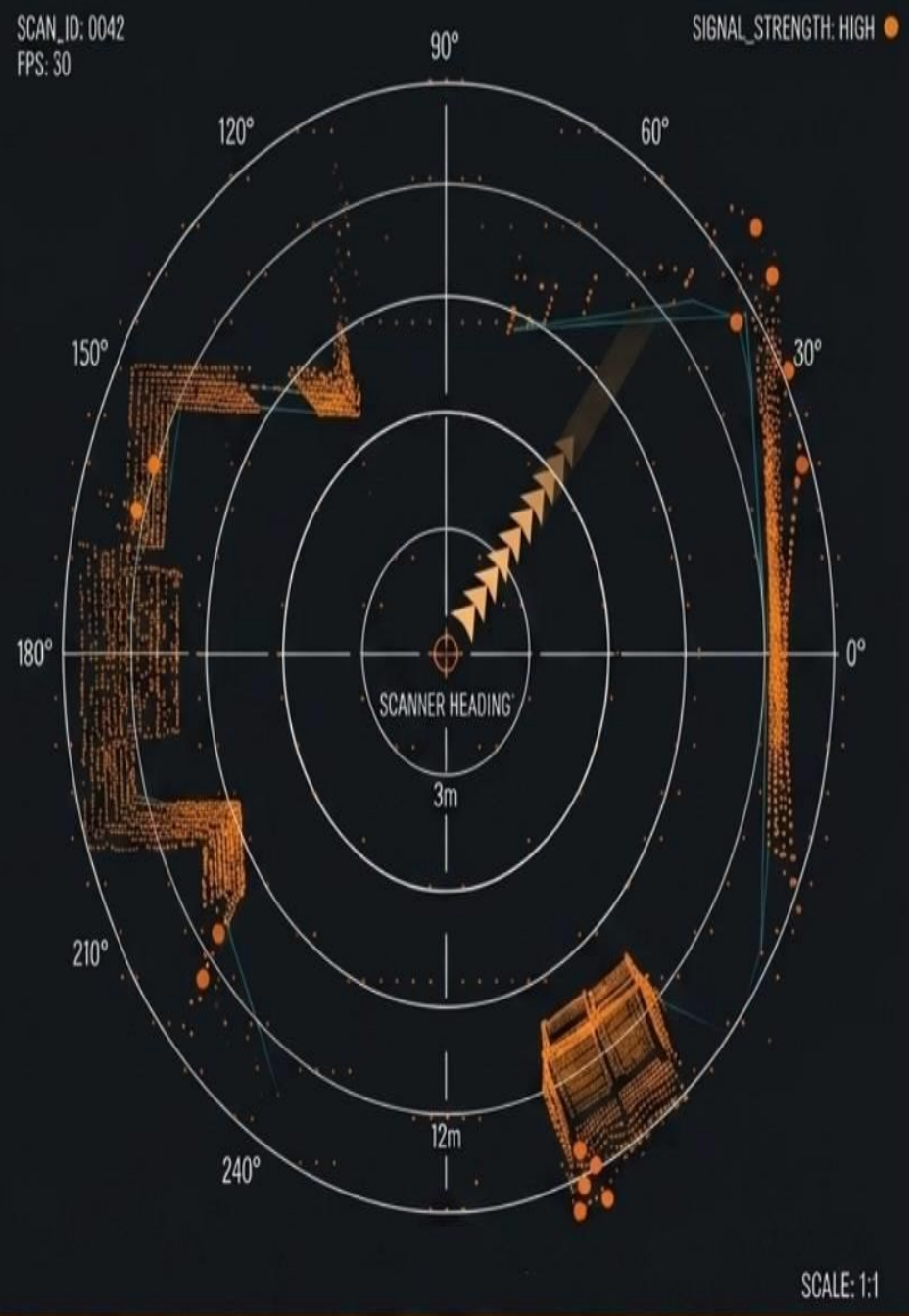
SLAMTEC

RPLiDAR

A1M8

360° Laser Range Scanner
Setup & Operation Guide



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1. Installing Python and VS Code

Python

Go to https://www.python.org/downloads and download the latest Python 3 installer,

You should see a version number such as Python 3.12.3 . download it , Run it and on the first screen check Add Python to PATH then click Install Now.

VS Code

Go to https://code.visualstudio.com and download the Windows installer. Run it and on the Select Additional Tasks screen check both Add to PATH and Open with Code. Once installed, open VS Code and install the Python extension: click the Extensions icon on the left sidebar, search Python, and install the one by Microsoft.

Creating a Python File

1. Open VS Code and click File Then Open Folder, then select or create a project folder.
2. In the Explorer panel click the New File icon and name the file lidar.py.
3. You can now write Python code in this file and run it from the built-in terminal (Terminal Then New Terminal).

2. Installing the Required Libraries

Open the VS Code terminal or Command Prompt and run the following command:

python -m pip install rplidar-roboticia pyserial matplotlib

Library	Purpose
rplidar-roboticia	Communicates with the RPLiDAR sensor over serial
pyserial	Handles COM port / serial communication on Windows
matplotlib	Used to plot scan data as a live polar chart

3. Connecting the RPLiDAR via USB

Physical Connection

Connect the RPLiDAR sensor to the included USB adapter board (CP2102) using the short cable in the box.

Plug the Micro-USB cable from the adapter board into any USB port on the computer.

Installing the USB Driver

The RPLiDAR communicates through a CP2102 USB-to-serial chip. Windows may not include the driver by default. To install it:

Go to the following URL and download the CP210x Windows Drivers zip file:

https://www.silabs.com/developers/usb-to-uart-bridge-vcp-drivers

Extract the zip and run the CP210xVCPInstaller_x64.exe installer inside, Unplug and re-plug the USB cable after the driver installs

Finding the COM Port

After connecting, Windows assigns the device a COM port number. To find it:

Press Win + X and open Device Manager.

Expand Ports (COM & LPT).

Look for Silicon Labs CP210x USB to UART Bridge the COM number in brackets (ex COM3) is what you will use in your code.

4. Running the Example Code

In VS Code open lidar.py and paste the code below. Change COM3 on line 2 to the COM port number found in Device Manager.

Ex

lidarpr.py > ...

```python
1  from rplidar import RPLidar
2  lidar = RPLidar('COM3')
3  print('Info:',   lidar.get_info())
4  print('Health:', lidar.get_health())
5  try:
6      for i, scan in enumerate(lidar.iter_scans()):
7          print(f'Scan {i + 1} — {len(scan)} measurements')
8          for quality, angle, distance in scan:
9              if quality > 0 and distance > 0:
10                 print(f'  Angle: {angle:.1f} deg   Distance: {distance:.1f} mm')
11         if i >= 4:   # Stop after 5 scans
12             break
13 except KeyboardInterrupt:
14     pass
15 finally:
16     lidar.stop()
17     lidar.disconnect()
18     print('Disconnected.')
19
```

Save the file, then in the VS Code terminal run:

python -m lidar.py

The LiDAR motor will spin and measurements will print to the terminal for 5 full rotations, then stop. Press Ctrl + C at any time to stop early.

Expected Output:

```
js> Info:   {'model': 40, 'firmware': (1, 29), 'hardware': 7}
Health: ('Good', 0)
Scan 1 — 380 measurements
  Angle: 0.9 deg   Distance: 843.0 mm
  Angle: 1.8 deg   Distance: 841.5 mm
  ...
```